

Agentic Graph Reasoning

Autonomous Knowledge Graph Navigation for Fact Verification and Hallucination Mitigation in Large Language Models

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Pre-defense

The problem

Language models answer factual questions **fluently**, and just as fluently when they do not hold the fact.

In our own no-retrieval control, on WebQSP:

- 661 entities asserted
- 179 of them **do not exist** in the knowledge graph
- that is 27.1%, stated with no hedge

The failure is not rare, and it is not visible in the answer.

The gap

Fluency and factuality are produced by the same mechanism, so the output gives the reader no signal about which one they are getting.

A system that is confidently wrong is worse than one that says it does not know.

Where existing approaches stop

Approach	What it does	Where it stops
Parametric	Answers from weights	No external evidence at all
Vector RAG	Retrieves text once, then reasons	Retrieval happens <i>before</i> reasoning begins
Static GraphRAG	Retrieves a fixed neighbourhood	Radius fixed before the question is understood
Agentic navigation	Interleaves retrieval and reasoning	Emits whatever the last generation call produces

Nobody checks what the answer asserts *before* it is delivered.

Research questions

- RQ1** Does agentic navigation improve multi-hop factual accuracy — and does the advantage **grow with hop count**?
- RQ2** Given a system that already navigates the graph, what does a claim-level check against traversed triples **contribute**?
- RQ3** Which components earn their cost, in accuracy and in **tokens**?

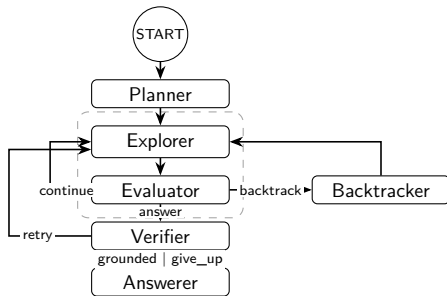
RQ2 is posed as *what does it contribute*, not *does it reduce hallucination*. The answer has several parts, and a yes/no framing would hide them.

AGR: an explicit state machine

Six nodes; the model is consulted only at decision points.

- **Planner** — sub-objectives
- **Explorer** — expands the frontier
- **Evaluator** — objective met?
- **Backtracker** — undo + ban list
- **Verifier** — the contribution
- **Answerer** — emits what survived

Two cycles, both bounded by budgets rather than by model behaviour.



Constrained tools, not free-form queries

Operation	Returns	Why it is capped
<code>get_relations</code>	Candidate relations	≤ 300 offered to the scorer
<code>get_neighbors</code>	Neighbours of an entity	≤ 200 per expansion
<code>verify_connection</code>	Adjacency in the full graph	Used only by the verifier
<code>link_entity</code>	Surface form $\rightarrow$ node	Three-stage resolver

The agent never writes Cypher, and every graph operation is logged with its arguments and result.

Determinism in the tools is what makes the verification layer possible.

The Structural Verification Layer

Before anything is emitted:

1. Draft answer is split into **atomic claims**
2. Each claim is checked against the **triples actually traversed**
3. Supported claims keep their supporting triples
4. Unsupported claims trigger **targeted re-exploration**, or are **dropped**

The system therefore **hedges rather than asserts**, and returns every answer with its evidence.

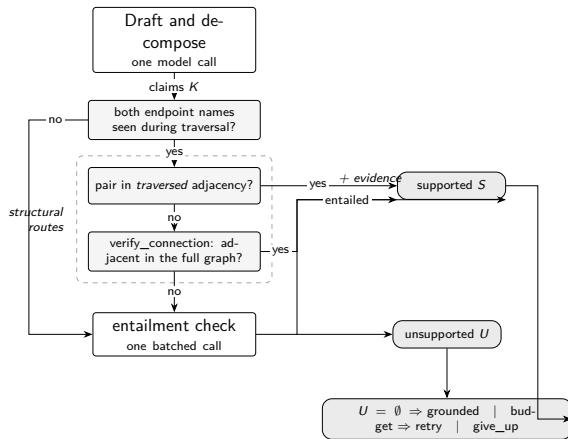
Why structural

The check is against the graph the agent walked — not against the model's own judgement of its own output. A model asked to grade itself is not an independent check.

The honest limit

A claim can be *true* and still be the wrong answer. Structural grounding cannot catch that — see the echo attractor later.

One claim, three routes



Only the third route costs a model call. The first two are pure graph lookups against the traversal record.

The environment and the question sets

Freebase-derived graph

Entities	2,592,892
Triples	8,309,194
Distinct relations	7,058
Import time	36.4 s

Certified question sets

	WebQSP	CWQ
Questions	400	400
Gold (median)	1.5	1.0
Reachable	97.0%	99.2%
Multi-hop	132	260

Reachability is the *ceiling* on every Hits@1 reported: a miss is the system's, not the environment's.

800 questions, with reachability certified per question so the accuracy ceiling is known.

Making the comparison fair

Five systems, **one frozen backbone, one budget:**

- No-retrieval (parametric control)
- Vector RAG
- Static GraphRAG
- Think-on-Graph (agentic)
- **AGR**

Same model, temperature 0, same 25-call cap, same questions, same graph.

What this buys

Differences are attributable to **architecture**, not to model capacity or to a bigger retrieval budget.

Why a parametric control

These benchmarks predate current models and may be partly memorised. Without the control we could not separate retrieval from recall.

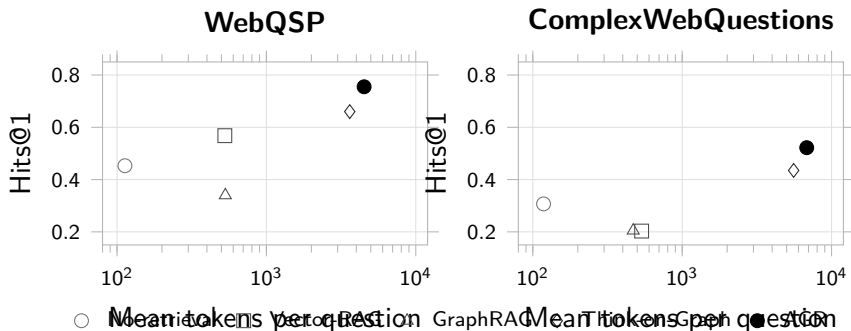
Main results

System	WebQSP		ComplexWebQuestions		Cost / question	
	Hits@1	F1	Hits@1	F1	Tokens	Calls
No-retrieval	0.453	0.305	0.307	0.279	113	1.0
Vector RAG	0.568	0.432	0.203	0.168	527	1.0
Static GraphRAG	0.340	0.262	0.205	0.183	531	1.0
Think-on-Graph	0.660	0.477	0.435	0.378	3,615	12.8
AGR	0.755	0.642	0.522	0.469	4,511	6.2

Cost columns are WebQSP. On CWQ, AGR spends 6,818 tokens and 8.9 calls against Think-on-Graph's 5,572 and 18.2.

Ahead of every baseline on both datasets, at half the model calls of the other agentic system.

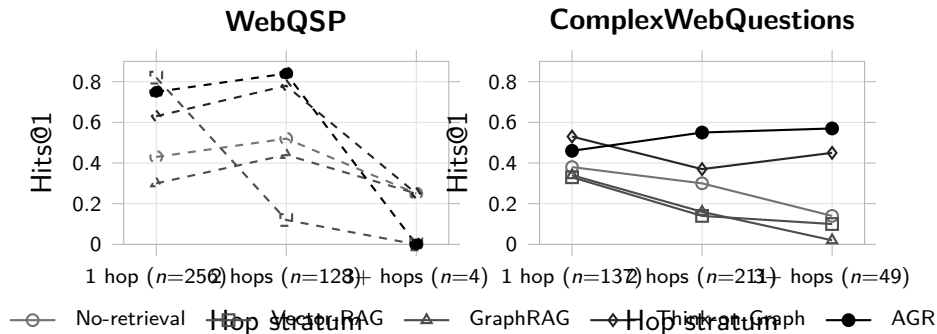
Accuracy against cost



On *tokens*, Think-on-Graph is **not** dominated — it buys lower accuracy at a genuinely lower token price. On *calls*, which is what the budget meters, it is dominated on both datasets.

The frontier depends on which cost you charge.

RQ1: accuracy against hop count



On ComplexWebQuestions AGR goes $0.46 \rightarrow 0.55 \rightarrow 0.57$ across one, two, and three-or-more hops — the **only** system that ends above where it started. Three of the other four decay monotonically.

RQ1 is answered by a shape, not a difference of means.

The caveat I want to raise myself

Hits@1 on...	WebQSP		ComplexWebQuestions	
	ToG	AGR	ToG	AGR
questions ToG <i>finishes</i>	0.852	0.788	0.629	0.607
questions ToG is <i>clipped</i> on	0.197	0.675	0.188	0.415
clip rate	29.2% (117/400)		44.0% (176/400)	

Where Think-on-Graph finishes inside the shared 25-call cap, it is **ahead of AGR on both datasets**. The entire aggregate margin comes from the questions it cannot finish. **The claim is efficiency under a fixed budget — not that AGR reasons better per step.**

RQ2: does anything ungrounded get asserted?

System	Entities asserted	Ungrounded	Rate
No-retrieval	1,001	221	22.1%
Vector RAG	1,329	45	3.4%
Static GraphRAG	1,024	8	0.8%
Think-on-Graph	1,265	0	0.0%
AGR	1,709	0	0.0%

Both datasets pooled. AGR asserts no entity that does not exist in the graph, across 1,709 assertions.

But Think-on-Graph reaches 0.0% too — so this is a property of *graph navigation*, not of the verification layer.

RQ2: so what does verification actually contribute?

What it does *not* do

Removing it changes accuracy by an amount we cannot detect: $p = 1.0$ on *both* datasets. It does not earn its place on Hits@1 or F1.

What it does do

- Turns silent error into a **hedge**: AGR hedges on 8.2% of WebQSP against no-retrieval's 12.2% error rate
- Attaches **supporting triples** to every asserted claim
- Gives an **auditable output contract**

This is a negative result reported as one. The layer's case rests on auditability, and the thesis says so rather than claiming an accuracy benefit the measurement does not support.

RQ3: what each component earns

Condition	WebQSP		ComplexWebQuestions		Tokens (WQ)
	F1	p	F1	p	
Full system	0.659	—	0.445	—	4,562
No planner	0.742	0.006	0.398	0.088	3,159
No backtracking	0.646	0.727	0.445	1.000	4,261
No verifier	0.663	1.000	0.436	1.000	4,460
Embedding-only scoring	0.643	0.664	0.437	0.481	3,413

Paired McNemar against the full system, stratified halves ($n = 200 / 198$).

Exactly one component shows a detectable accuracy effect — and the sign is backwards.

The result I did not expect

Removing the planner *improves* WebQSP:

- +0.083 F1, at $p = 0.006$
- -31% tokens
- 6.2 $\rightarrow$ 4.0 calls

On ComplexWebQuestions it trends the other way (-0.047 F1, $p = 0.088$) — not significant, but the opposite sign.

Why

WebQSP is predominantly one hop. Decomposing a one-hop question invents a second sub-objective that sends the frontier away from the answer.

What follows

Decomposition should be **gated on question structure**, not applied unconditionally. That is a design conclusion the measurement forced, not one it confirmed.

Every failure, read: the echo attractor

All 259 remaining failures read and labelled, both datasets pooled:

Category	n
Relation selection	65
Composite claim	47
Knowledge-graph gap	44
Decomposition error	38
Answer selection	15
Echo attractor	13

The characteristic error of a graph navigator is **not invention**. It is the **echo attractor**: a real, grounded entity one hop from the answer.

Why it matters

Any grounding check *passes* it — because it is true. Verification cannot catch it by construction.

It appears across systems, so it is a property of the task, not of AGR. Naming it is what lets future work target it.

The benchmark was wrong 57 times

Reading every failure also found questions where the *benchmark*, not the system, was at fault:

- 41 excluded before the census
- 17 found inside the census
- 1 counted in both

⇒ 57 **distinct questions**

A contribution that outlives this particular system.

Why report it

Reported defect rates for these two standard benchmarks are rare and usually anecdotal. This is a counted rate with a documented adjudication procedure and per-question provenance.

Contributions, and what I would not claim

Contributions

1. AGR: a state machine over a constrained graph tool API
2. The Structural Verification Layer and its output contract
3. A controlled five-system comparison, one backbone, one budget
4. The hop-count shape answering RQ1
5. The *echo attractor*, named
6. A counted benchmark-defect rate (57)

Limitations I state plainly

- Verification earns **no** detectable accuracy gain
- Zero ungrounded assertion is **navigation**, not the layer
- ToG leads on the questions it finishes
- One backbone: results are architectural, not universal

Thank you

Questions

Backup: budget configuration

Parameter	Value	Enforced in
max_depth	4	both routers
beam_width	3	explorer (slice)
max_anchors	5	explorer <i>and</i> evaluator
max_backtracks	3	post-evaluation router
max_llm_calls	25	LLM client
max_verify_iters	2	post-verification router, verifier
max_seconds	300	both routers

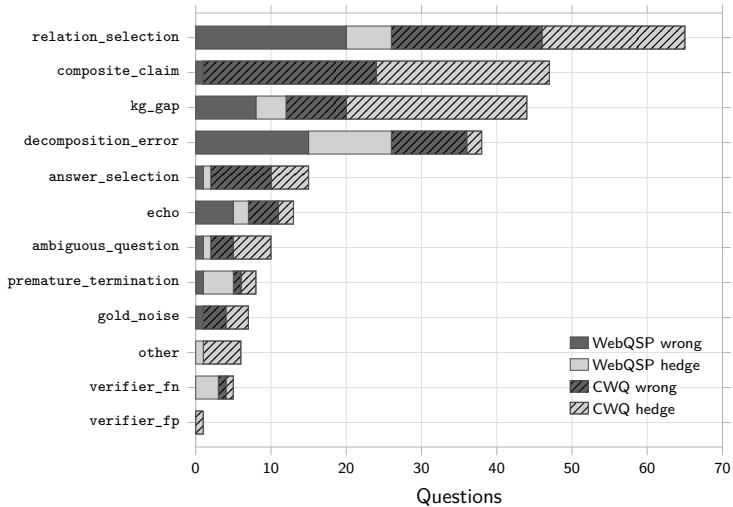
Termination does not depend on model behaviour: every cycle passes through a router that checks a monotone counter.

Backup: which budgets actually bind

Budget	WebQSP	CWQ	Pooled
Depth cap	16.5%	34.8%	25.6%
Backtrack cap	3.2%	9.2%	6.2%
Verify-iteration cap	1.0%	2.8%	1.9%
Call cap	0.0%	0.0%	0.0%

Percentage of questions on which each budget refused a request. AGR never reaches the 25-call cap — which is what makes the comparison against a clipped Think-on-Graph meaningful.

Backup: the full failure census



Backup: hedging behaviour

System	WebQSP hedge %	CWQ hedge %
No-retrieval	12.2	38.0
Vector RAG	27.0	48.2
Static GraphRAG	55.8	60.8
Think-on-Graph	18.8	22.5
AGR	8.2	23.0

A hedge is a refusal to assert, not a wrong answer. AGR hedges least on WebQSP while also being most accurate there.